SANDEEP BHAJANTRI

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PROFESSIONAL SUMMARY

*Third-year B.E. student in Artificial Intelligence & Machine Learning with hands-on experience building deep learning and computer vision systems in **PyTorch**, including **keypoint localization**, **image classification**, and **multi-agent LLM applications**. Skilled in model training, evaluation, and applied ML pipeline design, with national hackathon wins.*

EDUCATION

B.E. in Artificial Intelligence & Machine Learning — 2023 – 2027 (Expected)
B.M.S. Institute of Technology and Management, Bengaluru

Pre-University Course (PUC), PCMC — 2023 | 83.17% (499/600)
Morarji Desai Residential PU College, kalli, Karnataka

SSLC — 2021 | 84.96% (531/625)
Ave Maria High School, Sirsi, Karnataka

TECHNICAL SKILLS

Programming: **Python**

Machine Learning: **Supervised & Unsupervised Learning, Transfer Learning, Model Evaluation, Data Preprocessing, Class Imbalance Handling, Data Augmentation**

Deep Learning**: PyTorch, TensorFlow, Scikit-learn, CNNs, ResNet**

Computer Vision**: OpenCV, Object Detection (YOLOv8), Keypoint Localization, Pose Estimation, Image Classification, Image Restoration (GANs, DnCNN)**

NLP / Generative AI: **LLMs, Hugging Face Transformers, RAG, LangChain, Prompt Engineering, Agentic AI, Multi-Agent Systems, LLM Agents, Tool Calling, Agent Orchestration**

AI Engineering: **Agent Orchestration, Retrieval Pipelines, Model Inference, FastAPI, Flask, Gradio, REST APIs, Docker**

Tools: **Git, GitHub, Google Colab, Jupyter, Weights & Biases, FFmpeg**

PROJECTS

Aerial GCP Pose Estimation & Shape Classification

Computer Vision • PyTorch • ResNet18 • Keypoint Localization

- Built a computer vision pipeline in **PyTorch** with a **ResNet18** backbone to jointly localize Ground Control Point **(GCP)** keypoints and classify **GCP shapes** (Cross, Square, L-Shape) in **aerial drone imagery**.
- Curated and preprocessed a dataset of **996 annotated aerial images**, applying image resizing and **coordinate normalization** to standardize inputs for keypoint **regression and classification**.
- Designed a **multi-task** loss combining **coordinate regression loss with cross-entropy classification loss**, applying weighted loss terms to address class imbalance across GCP shape categories.
- Implemented train/validation splitting and batch training using **PyTorch DataLoader** and the **AdamW** optimizer, evaluating model performance through prediction vs. ground-truth comparison.

Multi-Agent AI System

Agentic AI • LLMs • Python • Retrieval-Augmented Generation

- Built a **multi-agent AI** system in Python where coordinated **LLM-based** agents interpreted natural language queries and retrieved relevant records from structured client data.
- Designed **agent coordination logic to route incoming queries**, retrieve applicable client records, and generate grounded responses using **retrieval-augmented** techniques.
- Implemented reasoning and **tool-calling patterns** enabling agents to invoke retrieval tools and compose responses grounded in retrieved record data.
- Tested the **multi-agent pipeline** against varied client-record queries to evaluate response reliability and consistency.

ACHIEVEMENTS

- **Winner** — National-Level Hackathon, IIIT Gwalior (2025) | Competed against **500+ teams** nationwide
- **Top 4** — NMAM Hackfest 2025
- **Finalist** — Hack2Future 2024, IIIT Dharwad
- **Participant — International BizThon Hackathon, in collaboration with Yellow Network, IIT Delhi (2025)**